

SHENKURSK, Russian Federation

WMO: 227680

Lat: 62.1140N

Lon: 42.8899E

Elev: 147

StdP: 14.62

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-19.7	-30.8	1.0	-25.7	-24.8	1.4	-19.8	18.7	20.8	17.3	21.3	2.3	190	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	83.6	66.3	79.9	64.7	76.3	62.8	68.8	79.2	66.7	76.4	64.8	73.5	7.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.0	93.1	73.2	63.1	87.0	71.2	61.1	81.0	69.4	33.0	78.9	31.4	76.7	29.9	73.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.6	13.8	DB	-33.0	88.1	6.4	3.3	-37.6	90.4	-41.4	92.4	-45.0	94.2	-49.6	96.7	(4)
(5)			WB	-33.2	71.3	6.4	2.5	-37.8	73.1	-41.5	74.6	-45.1	76.0	-49.7	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	10.2	11.8	22.7	36.9	48.8	58.2	64.4	58.6	49.2	37.7	24.2	16.2		
	DBStd	21.38	15.95	15.22	10.19	8.93	9.59	8.26	7.40	7.18	6.84	7.45	11.75	14.63		
	HDD50	6029	1235	1069	846	402	144	18	2	8	95	387	775	1048		
	HDD65	10475	1700	1489	1311	842	507	238	104	225	474	847	1225	1513		
	CDD50	1180	0	0	0	11	107	262	449	274	72	5	0	0		
	CDD65	150	0	0	0	0	5	33	86	26	0	0	0	0		
	CDH74	1276	0	0	0	2	55	318	705	194	2	0	0	0		
	CDH80	311	0	0	0	0	7	85	170	49	0	0	0	0		
(14) Wind	WSAvg	6.7	6.9	6.9	7.1	6.7	7.0	6.5	5.8	5.7	6.2	7.1	7.0	7.3		
(15) Precipitation	PrecAvg	22.4	1.3	1.1	1.1	1.5	2.0	2.5	2.7	3.0	2.1	2.0	1.6	1.6		
	PrecMax	29.4	2.6	2.0	3.2	4.6	3.4	6.1	6.3	7.1	4.4	3.7	2.8	3.7		
	PrecMin	15.2	0.3	0.0	0.2	0.2	0.6	0.6	0.8	0.5	0.6	0.0	0.5	0.6		
	PrecStd	3.0	0.5	0.6	0.7	0.9	0.9	1.3	1.4	1.5	0.9	0.7	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	36.7	47.1	69.1	80.7	87.2	88.6	85.8	72.4	60.1	45.9	40.1		
		MCWB	34.4	34.7	37.6	53.8	59.9	66.7	69.2	68.6	60.4	52.7	43.5	38.6		
	2%	DB	34.1	34.7	40.9	61.7	74.2	82.1	84.4	79.3	66.6	54.3	41.7	36.3		
		MCWB	33.0	33.0	34.8	48.7	57.0	64.5	67.2	65.2	58.1	50.1	39.8	34.7		
	5%	DB	32.7	33.2	37.6	54.5	69.8	77.2	81.1	74.3	62.8	50.8	39.1	34.6		
		MCWB	31.6	31.8	33.5	44.4	54.5	62.0	65.9	62.9	56.1	47.8	37.9	33.2		
	10%	DB	30.2	31.1	35.6	50.1	65.1	73.0	77.9	70.6	59.5	48.2	36.4	33.1		
		MCWB	29.1	29.8	32.7	41.6	52.4	59.9	64.5	61.1	54.5	45.9	35.1	31.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	35.1	39.0	54.5	63.4	69.4	72.1	70.8	61.7	54.1	43.9	38.5		
		MCDB	35.5	36.3	45.2	68.3	74.4	82.2	83.0	81.8	68.0	58.0	45.4	39.4		
	2%	WB	33.1	33.3	36.1	49.6	59.4	66.2	69.5	67.1	59.2	50.9	40.1	35.1		
		MCDB	34.0	34.3	39.3	58.8	69.9	78.7	80.3	75.7	64.5	53.6	41.5	35.9		
	5%	WB	31.7	31.9	34.5	45.6	56.6	64.1	67.7	64.5	56.9	48.2	37.9	33.3		
		MCDB	32.8	33.2	36.9	53.6	66.0	73.8	77.3	72.3	61.7	50.5	39.1	34.2		
	10%	WB	29.0	29.8	33.0	42.2	54.0	61.7	66.1	62.2	55.0	45.7	35.2	32.0		
		MCDB	30.1	31.1	35.4	48.9	63.7	70.6	75.6	68.5	59.3	48.0	36.2	33.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.5	14.9	15.3	16.7	16.9	16.6	14.7	12.1	7.5	7.3	9.2		
		MCDBR	9.5	8.5	14.5	22.5	23.7	22.8	21.7	20.2	17.5	11.6	7.2	6.5		
	5% WB	MCWBR	9.1	8.0	10.3	12.5	10.4	9.5	8.2	8.5	9.9	8.2	6.7	6.4		
		MCWBR	8.9	8.2	8.0	12.2	10.0	9.6	8.3	8.4	9.6	8.4	6.8	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.297	0.325	0.364	0.354	0.353	0.391	0.371	0.338	0.317	0.266	0.220		
	taud		1.999	2.065	2.093	2.198	2.438	2.469	2.373	2.404	2.484	2.377	2.107	1.956		
	Ebn at Noon		129	201	242	259	275	277	261	257	245	202	122	79		
	Edh at Noon		12	23	33	36	31	31	33	30	23	17	11	7		
(44) All-Sky Solar Radiation	RadAvg		77	284	718	1207	1552	1663	1646	1152	649	247	72	35		
	RadStd		10	64	114	124	100	180	135	120	76	34	15	5		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	+1.30	N/A	N/A	N/A	N/A	-400	-455	N/A	N/A	(47)

Nomenclature: See separate page